

Math 252, First Examination Review

Directions: Respond to each question in the space provided, under each question. You may use the back of the page if necessary. All calculations and mathematical work must be shown and justified, including citation of well known Theorems or properties that have been used for solving the problem. Only a TI-30X (type) scientific calculator may be used on this exam. This is a closed note, closed book exam. No notes, laptops, smartwatches, mobile devices, or any other unapproved electronic device may be used while taking this exam.

1. Consider the graphs of $y = 4 - x^2$ and $y = x + 2$.
 - (a) Sketch a graph of the functions and determine x -values at which the graphs intersect.
 - (b) Let R be the region bounded between the two graphs. Write **AND** evaluate an integral for the area of R .
2. Let $f(x) = x^3 - x$ and $g(x) = x - x^2$.
 - (a) Sketch a graph of the functions and determine x -values at which the graphs intersect.
 - (b) Let R be the region bounded between the two graphs. Write **AND** evaluate an integral for the area of R .
3. Let D be the region bounded between the graph of $y = x^2 + 1$ for $0 \leq x \leq 3$.
 - (a) Sketch a graph of the region clearly indicating the region D , and set up an integral for the volume of the solid obtained by revolving D around the line $y = 10$. DO NOT EVALUATE THE INTEGRAL
 - (b) Set up an integral for the volume of the solid obtained by revolving D around the line $x = -1$. DO NOT EVALUATE THE INTEGRAL.
4. Let R be the region bounded between the graphs of $f(x) = x - 1$ and $g(x) = x^2 - 1$.
 - (a) Sketch a graph of the region clearly indicating the region R , and set up an integral for the volume of the solid obtained by revolving R around the line $y = -2$. DO NOT EVALUATE THE INTEGRAL
 - (b) Set up an integral for the volume of the solid obtained by revolving R around the line $x = 2$.
5. The base of a solid is bounded between the positive y -axis, the graph of $y = 4 - x^2$, and the negative x -axis for $-2 \leq x \leq 0$, and whose cross-sections are squares with a side that runs perpendicular to the x -axis.
 - (a) Sketch a graph of the base, and another picture of the cross-section.
 - (b) Set up an integral for the volume of the solid. DO NOT EVALUATE IT. Note: $A_{\text{square}} = s^2$.
6. The base of a solid is the disk of radius 2 (i.e. $x^2 + y^2 = 4$).
 - (a) Sketch a graph of the base, and another picture of the cross-section.

- (b) If the cross-sections are isosceles triangles, perpendicular to the x -axis, write an integral for the volume of the solid. DO NOT evaluate the integral. ($A_{eq. triangle} = \frac{1}{2}s^2$)
7. We have looked at a lot of Riemann Sums during this unit. Consider the following Riemann Sum, $\sum_{i=1}^n \left(\frac{2i}{n} \sqrt{1 + \frac{2i}{n}} \right) \cdot \frac{2}{n}$, where $0 \leq x \leq 2$.
- (a) What is the Δx and determine if left end point rule, right endpoint rule, or midpoint rule was used.
- (b) Complete the following statement: The Riemann Sum will converge to the integral _____, when _____ (DO NOT EVALUATE THE INTEGRAL OR THE SUM).
8. We have looked at a lot of Riemann Sums during this unit. Consider the following Riemann Sum, $\sum_{i=1}^n \frac{(2k-1)\pi}{2n} \sin^2 \left(\frac{(2k-1)\pi}{2n} \right) \cdot \frac{\pi}{n}$, where $0 \leq x \leq \pi$.
- (a) What is the Δx and determine if left end point rule, right endpoint rule, or midpoint rule was used?
- (b) Complete the following statement: The Riemann Sum will converge to the integral _____, when _____ (DO NOT EVALUATE THE INTEGRAL OR THE SUM).
9. (a) Using substitution, evaluate $\int \frac{1}{\sqrt{1+4x}} dx$.
- (b) Evaluate $\int \sec 3x dx$.
10. (a) Let $f(x) = \frac{1}{2}(e^x + e^{-x})$ for $0 \leq x \leq 1$. Determine the arc length of the graph (leave your answer with e in it).
- (b) Let $f(x) = \frac{4}{3}x^{3/2}$ for $1 \leq x \leq 2$. Determine the arc length of the graph (leave your answer in its exact form...NO DECIMALS). *Hint: Use part (a)*